9th Australian Algebra Conference

November 17 – 18, 2025

La Trobe University

Melbourne Australia

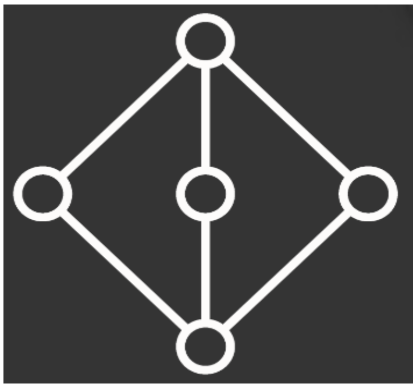
The 9th Australian Algebra Conference is sponsored by the LTU School of Computing, Engineering and Mathematics and the Australian Algebra Group.

We acknowledge and pay respect to the Traditional Owners and Elders – past, present and emerging – of the lands on which La Trobe University operates.

General Information

Goal and Topics

The main aim of the annual Australian Algebra Conference is to foster communication between algebraists in Australia. We interpret algebra quite broadly, including areas such as topological algebra, algebraic logic, graph theory and coding theory. The conference is run by the Australian Algebra Group, which is a special interest group of the Australian Mathematical Society.



Invited Speakers

P. A. Azeef Muhammed, *Western Sydney University*.

Nik Ruškuc, *University of St Andrews*.

Adam Thomas, *University of Warwick*.

Organising Committee

Santi Barrera Acevedo, *La Trobe University*.

Marcel Jackson, *La Trobe University*.

Olga Minchin, *La Trobe University*.

Local Information

Where: The conference will be held at La Trobe University City Campus – Levels 2 – 3 / 360 Collins St, Melbourne VIC 3000.

Talks on Monday and Tuesday will be held in rooms 3.08 and 3.09 Level 3.

Lifts: From the foyer, take the lifts serving floors 1 – 11. Make sure to select Level 3 using the black panels located outside the lifts, and proceed to the lift indicated on the screen. **You cannot select a floor from inside the lift itself**.

Internet access: Eduroam is available.

Toilet facilities: On Level 3, both gendered and gender-neutral toilets are located in the lift foyer area. You can also use the toilet facilities on Level 2.

Coffee and Food

Simple tea and coffee will be offered during the morning and afternoon breaks.

There is a café called Code Black on the ground floor, they offer a variety of coffees, teas and pastries.

At the back of the building on **Little Collins Street**, there is a small food hub with several cafes and chains, including Earl, Fishbowl, Yummy Mono, Greenstreat, and Tomo's.

Another food hub is **Flinders Lane** and surroundings (6 minute walk). We particularly recommend Brunetti Oro for sandwiches, paninis and pizzas. For great coffee (especially cold brew), we recommend Dukes Coffee Roasters. As for great matcha, we recommend Miyama in Melbourne Central (10 minute walk).

Australian Algebra Group Annual General Meeting

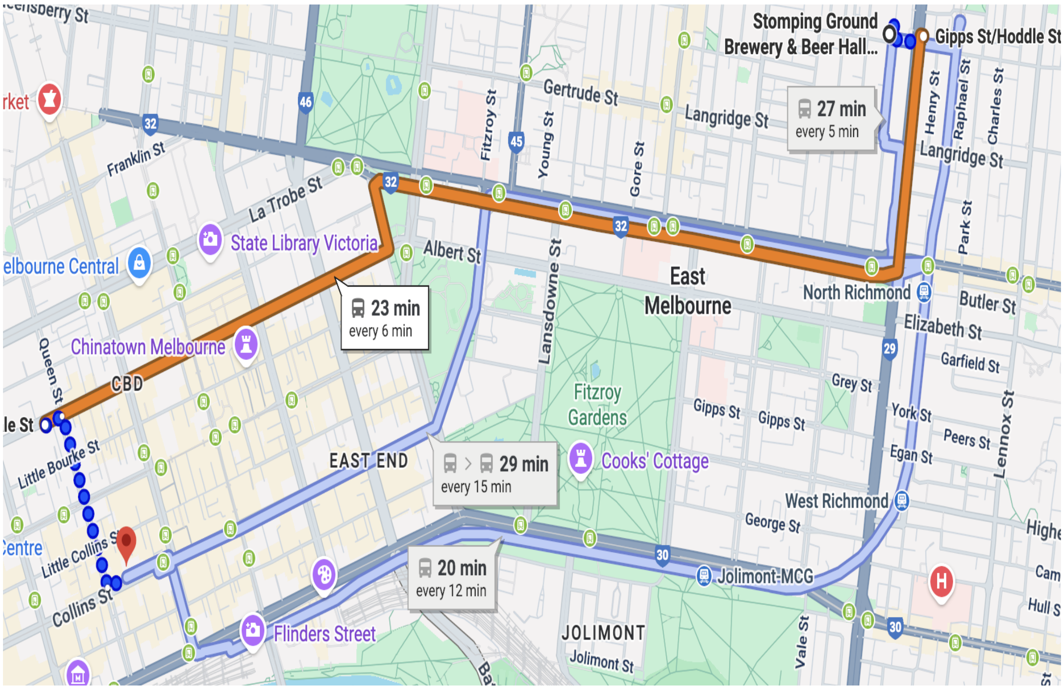
The annual general meeting for the Australian Algebra Group will take place on Monday 17, from 12:30 pm to 1:30 pm in room 2.11 Level 2. If you are attending this meeting, we suggest grabbing a quick bite or coffee from the café on the ground level or from one of the nearby bakeries.

Conference Dinner

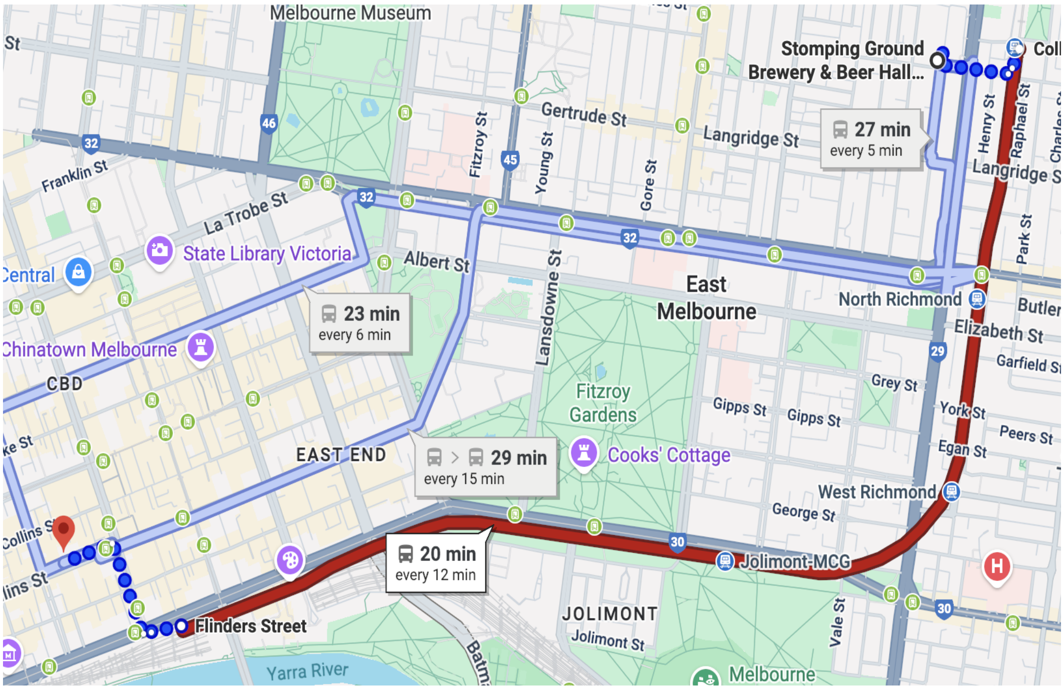
The (self-funded) conference dinner will be held at Stomping Ground Brewery (100 Gipps St, **Collingwood** 3066) on Monday, November 17, from 5:30 pm to 8:30 pm. Stomping Ground offers a relaxed, welcoming atmosphere and has earned multiple awards, including Champion Medium Australian Brewery (AIBA 2024 & 2022) and

Champion Australian Independent Large Brewery (Indies 2021). **There is happy hour until 6pm**. There are also $10 pints for main beers. Here is the menu. On average it takes about **30 minutes to commute** from LTU to Stomping Ground.

Bus option



Train option



Uber: It costs roughly $10 and is cheaper than public transport if you carpool.

List of Participants

P.A. Azeef Muhammed, Western Sydney University
Reymond Akpanya, The University of Sydney
Brice Arrigo, University of Melbourne
Santi Barrera Acevedo, La Trobe University
Michal Botur, Palacký University Olomouc
Luka Carroll, Western Sydney University
Greta Civani, The University of Melbourne
Graham Clarke, RMIT University
Brian Davey, La Trobe University
Heiko Dietrich, Monash University
Qian Ding, La Trobe University
James East, Western Sydney University
Murray Elder, University of Technology Sydney
Alex Elzenaar, Monash University
Fawwaz Fakhrurrozi, The University of Melbourne
Alexander Fish, The University of Sydney
Matthias Fresacher, Western Sydney University
Barry Gardner, University of Tasmania
Stephen Glasby, The University of Western Australia
Yusuf Hafidh, The University of Melbourne
Joshua Howie, Monash University
Clive Jackson,
Deborah Jackson, La Trobe University
Marcel Jackson, La Trobe University
Seethalakshmi Kayanattath, Australian National University
Finn Klein, The University of Sydney
Melissa Lee, Monash University
JS Lemay, Macquarie University,
Shir Levav-Porat, Monash University
Mengfan Lyu, Western Sydney University
Lavender Marshall, Monash University
Eileen Pan, Monash University
Phoenix Pham, The University of Melbourne
Anthony Pisani, Monash University
Tomasz Popiel, Monash University
John Power, Macquarie University

Devi Imulia Dian Primaskun, The University of Melbourne
Ganesha Lapenangga Putra, The University of Melbourne
Tao Qin, The University of Sydney
Nik Ruškuc, University of St Andrews
Snehinh Sen, Australian National University
Joshua Tan, University of Technology Sydney
Don Taylor, University of Sydney,
Adam Thomas, University of Warwick
Vandit Trivedi, Australian National University
Stefan Veldsman, Nelson Mandela University and La Trobe University
Joshua Walters, The University of Melbourne
Yu Wang, The University of Melbourne
Vindya Warnakulasooriyage, RMIT University
Bailey Whitbread, The University of Sydney
Jung Won Cho, University of St Andrews
Binzhou Xia, The University of Melbourne

Schedule

	Monday	Session A Room 3.08	Session B Room 3.09
9:00 - 9:10	Conference opening		
9:10 - 10:10	Invited talk	Nik Ruškuc	
10:10 - 10:35	Talk	James East	Reymond Akpanya
10:35 - 10:55	**Coffee break**		
10:55 - 11:20	Talk	Matthias Fresacher	Vindya Warnakula-sooriyage
11:20 - 11:45	Talk	Jung Won Cho	Anthony Pisani
11:45 - 12:10	Talk	Luka Carroll	Vandit Trivedi
12:10 - 1:40	**Lunch / AGM (12:30–1:30)**		
1:40 - 2:40	Invited talk	P. A. Azeef Muhammed	
2:40 - 3:05	Talk	Marcel Jackson	Binzhou Xia
3:05 - 3:30	Talk	Snehinh Sen	Yusuf Hafidh
3:30 - 3:50	**Coffee break**		
3:50 - 4:15	Talk	JS Lemay	Melissa Lee
4:15 - 4:40	Talk	Qian Ding	Bailey Whitbread

	Tuesday	Session A Room 3.08	Session B Room 3.09
9:00 - 10:00	Invited talk	Adam Thomas	
10:00 - 10:25	Talk	Heiko Dietrich	Mengfan Lyu
10:25 - 10:45	**Coffee break**		
10:45 - 11:10	Talk	Lavender Marshall	Tao Qin
11:10 - 11:35	Talk	Alex Elzenaar	Seethalakshmi Kayanattath
11:35 - 12:00	Talk	Mike Zhang	Finn Klein
12:00 - 1:30	**Lunch**		
1:30 - 1:55	Talk	Michal Botur	Barry Gardner
1:55 - 2:20	Talk	John Power	Alexander Fish
2:20 - 2:45	Talk	Don Taylor	Stefan Veldsman
2:45 - 3:10	Talk	Stephen Glasby	Joshua Howie
3:10 - 3:25	Closing remarks	All	

Book of Abstracts

Monday 17 of November

Session A - Room 3.08

Automorphism groups of filtered Boolean powers

Nik Ruškuc
University of St Andrews

Let A be a finite non-abelian simple algebra in a congruence permutable variety; the latter include groups, rings, associative and Lie algebras, loops and quasigroups. Let B be a Boolean algebra. We discuss the automorphism group G of the Boolean power of A by B, consisting of all continuous functions from the Stone space X of B into A; or, more generally, of the filtered version, where values are fixed for finitely many points in X. We exhibit a general structure theorem for G as a certain semidirect product. As a consequence, in the case where B is the countable atomless Boolean algebra, we obtain several attractive properties for G, generally associated with Fraisse limits: small index property, Bergman property, ample generics and Ramsey property. Most of the latter are new even in the classical settings of groups and rings. This is joint work with Peter Mayr from the Colorado University at Boulder.

Matrix representations of diagram monoids

James East
Western Sydney University

The partition monoid P_n is an important algebraic object that arises in many parts of mathematics and science. It contains well known submonoids such as the Brauer monoid B_n and the Temperley–Lieb monoid TL_n. We show that P_n has a faithful 2^n-dimensional representation by zero-one matrices over any additively-idempotent semiring. In the case of the Boolean semiring, this is equivalent to a representation by binary relations. The restrictions to B_n and TL_n contain faithful subrepresentations of lower dimension; for TL_n the dimension is the nth Fibonacci number. Interpreting the zero-one matrices instead over a ring of appropriate characteristic, one obtains faithful representations of the (finite or infinite) twisted versions of these monoids. This is all joint work with Marianne Johnson and Mark Kambites (Univ Manchester).

Categorical representation of DRC-semigroups

Matthias Fresacher
Western Sydney University

DRC-semigroups model associative systems with domain and range operations, and contain many important classes, such as inverse, restriction, Ehresmann, regular $*$-, and $*$-regular semigroups. In this joint work with James East, Azeef Muhammed and Tim Stokes, we show that the category of DRC-semigroups is isomorphic to a category of certain biordered categories whose object sets are projection algebras in the sense of Jones. This extends the recent groupoid approach to regular $*$-semigroups of East and Muhammed. We also establish the existence of free DRC-semigroups by constructing a left adjoint to the forgetful functor into the category of projection algebras.

Free inverse semigroups and the Howson property

Jung Won Cho
University of St Andrews

The elements of a free inverse semigroup can be represented by directed, birooted, edge-labelled graphs called Munn trees. In this talk, I will introduce the Munn trees and explore, in particular, the monogenic (i.e. generated by a single element) free inverse semigroup. I will then discuss the Howson property: an algebra has the Howson property if the intersection of two finitely generated subalgebras is again finitely generated. We will see that the monogenic free inverse semigroup is the only monogenic inverse semigroup which does not have the (semigroup) Howson property. This is joint work with Craig Miller and Nik Ruškuc.

Presentations for semigroups of full-domain partitions and trivial-cokernel partitions

Luka Carroll
Western Sydney University

East and Gray in their 2021 paper uncovered several interesting substructures of the partition monoid $\mathcal{P}_n$ that arise from natural unary operations. The most notable of these structures include the full-domain partition monoid $\mathcal{P}_n^{\mathrm{fd}}$ and the trivial-cokernel partition monoid $\mathcal{P}_n^{\mathrm{tc}}$. Both $\mathcal{P}_n^{\mathrm{fd}}$ and $\mathcal{P}_n^{\mathrm{tc}}$ extend the full transformation monoid $\mathcal{T}_n$ on $\mathbf{n} = \{1, 2, \ldots, n\}$, the set of all mappings $\mathbf{n} \to \mathbf{n}$ under composition, by appropriate semilattices. This is analagous to how the partial transformation monoid $\mathcal{PT}_n$ extends $\mathcal{T}_n$ by the semilattice of partial identities. The monoid $\mathcal{P}_n^{\mathrm{fd}}$ extends $\mathcal{T}_n$ by $(\mathfrak{Eq}_n, \vee)$, the join semilattice of equivalences on $\mathbf{n}$, whilst $\mathcal{P}_n^{\mathrm{tc}}$ extends $\mathcal{T}_n$ via $(2^{\mathbf{n}}, \wedge)$, the meet semilattice of subsets of $\mathbf{n}$.

This talk will discuss recent work that has been taken to initiate the study of $\mathcal{P}_n^{\mathrm{fd}}$ and $\mathcal{P}_n^{\mathrm{tc}}$ by finding presentations (by generators and relations) of the monoids and their own substructures, including some intriguing $\mathcal{L}$- and $\mathcal{R}$-trivial submonoids (where $\mathcal{L}$ and $\mathcal{R}$ describe Green's left and right divisibility relations). Whilst many of these presentations have been proven directly, some have used generalised techniques (developed by Carson, Dolinka, East, Gould and Zenab in their 2023 paper) for handling restriction monoids such as $\mathcal{P}_n^{\mathrm{fd}}$.

This is joint work with James East, Matthias Fresacher, Mengfan Lyu and Azeef Muhammed.

Partial symmetry, partitions and matrices

P. A. Azeef Muhammed
Western Sydney University

Just as groups are algebraic models of symmetry, inverse semigroups capture the notion of partial symmetry. One of the major results in algebraic semigroup theory is the ESN (Ehresmann-Schein-Nambooripad) Theorem which describes the structure of inverse semigroups using certain groupoids. In this talk, we discuss how we can generalise this to the class of regular $*$-semigroups, and further to $*$-regular semigroups. Just like inverse semigroups, both these classes have an involution to capture the symmetry, but their idempotents no longer form a semilattice. Natural examples of regular $*$-semigroups include the partition, Brauer and Temperley-Lieb monoids among other diagram monoids. Every regular $*$-semigroup is indeed $*$-regular but not conversely. The prototypical example of a $*$-regular semigroup is the full matrix monoid over $\mathbb{R}$ or $\mathbb{C}$ with respect to Moore-Penrose inverses. By focussing on the *projection algebra* structure of these classes, we obtain a category isomorphism between these classes of semigroups and certain ordered groupoids, just as in the ESN theorem. This is a recent joint work with James East.

Additively idempotent semirings via block hypergraphs

Marcel Jackson
La Trobe University

The study of additively idempotent semirings (also known as semilattice-ordered semigroups) has a decidably combinatorial feel, with a number of known encodings of combinatorial objects into equational problems. This talk continues the theme by considering the variety (meaning equational class) generated by the smallest challenging semiring: the 3-element semiring known as "S_7". We explain how the subdirectly irreducible members of this variety are in a meaningful one-to-one correspondence with the class of hypergraphs we call "block hypergraphs", and which are reminiscent of both the Kneser graph family and the line graph construction. In fact, Kneser hypergraphs are examples of block hypergraphs, and we can use the rather famous homomorphism properties of Kneser graphs to establish that the lattice of subvarieties of the variety generated by S_7 has size of the continuum, embedding a chain isomorphic to the usual order on the real numbers.

The Canonical Positive Model

Snehinh Sen
The Australian National University

This talk would discuss some aspects of the intriguing and evolving theory of semirings. In short, a semiring is a "ring possibly without subtraction". So we allow $\mathbb{N}, \mathbb{R}_{\geq 0}$ and the Booleans $\mathbb{B}$. More generally, for any ring homomorphism, $f : R \to \mathbb{R}$, the *canonical positive model* $S_f = f^{-1}[0, \infty)$ is a semiring. For example, when R is a subring of $\mathbb{R}$, we may look at $S = R_{\geq 0}$. The main goal of this talk would be to describe and prove certain finiteness and flatness properties (following [1], [2], [3]) for the canonical positive model $R_{\geq 0}$. We would also talk about some general results on a seemingly unexplored notion of finiteness, which we call *Congruence Noetherian*, for semirings. This is a part of an ongoing research project with my PhD supervisor, Professor James Borger.

[1] James Borger. Witt vectors, semirings, and total positivity. *Absolute arithmetic and F_1−geometry*, pages 273–329. Eur. Math. Soc., Zürich, 2016.

[2] James Borger and Jaiung Jun. Facets of module theory over semirings, 2025.

[3] Yefim Katsov. On flat semimodules over semirings. *Algebra Universalis*, 51(2-3):287–299, 2004.

Dagger-Drazin Inverses

Jean-Simon Pacaud Lemay
Macquarie University

Drazin inverses are a special kind of generalized inverse that have been extensively studied and have many applications in ring theory, semigroup theory, and matrix theory. Drazin inverses can also be defined for endomorphisms in any category. A natural question is whether one can extend the notion of Drazin inverse to arbitrary maps - not just endomorphisms. It turns out that this is natural to do in dagger categories, which are multiobject analogue of involutive semigroups/monoids. In this talk I will introduce dagger Drazin inverses, a new kind of generalised inverse for maps in dagger categories, or simply elements of an involutive semigroup. I will also explain how dagger Drazin inverses are deeply connected to Drazin inverses and Moore-Penrose inverses. This is based on joint work with Robin Cockett and Priyaa Varshinee Srinivasan, and our paper: https://arxiv.org/abs/2502.05306v3

Universal axiomatizability in finite unary relational structures

Qian Ding
La Trobe University

The Łos-Tarski Theorem states that a class of models of a first order sentence is closed under taking substructures if and only if the sentence is equivalent to a universal sentence: a sentence in which all quantifiers (when moved to the front of the sentence) are universal.

It is known that the Łos-Tarski Theorem fails when restricted to finite models only. In this talk we explain why the Łos-Tarski Theorem is recovered on finite models if restricted to finite unary relation structures, though it fails on any signature containing a non-unary relation.

Monday 17 of November

Session B - Room 3.09

Simplicial Surfaces with given Automorphism Group

Reymond Akpanya
The University of Sydney

Frucht showed that every finite group can be realised as the automorphism group of a cubic graph. In this talk, we simplify Frucht's construction for groups generated by two elements, resulting in cubic graphs with fewer vertices. In the general case, we identify an oversight in Frucht's original construction and present a modified construction that yields cubic graphs whose automorphism groups are isomorphic to the given groups. Additionally, we show that these graphs admit cycle double covers, leading to simplicial surfaces with automorphism groups that are isomorphic to prescribed finite groups. This is joint work with Tom Goertzen.

Characterisation of Latin Squares Induced by Bipermutive Cellular Automata

Vindya Warnakulasooriyage
RMIT University

A Latin square is an $n \times n$ square matrix in which each of the numbers $1, 2, \ldots, n$ appears exactly once in each row and each column. This research aims to characterise Latin squares generated by cellular automata (CAs) by investigating their structural and algebraic properties. A discrete dynamical system described by a regular lattice of cells is a cellular automaton (CA), where each cell synchronously updates its value by applying a local rule to itself and its neighbouring cells. We establish that the necessary and sufficient condition for a CA to generate a Latin square is that the CA is bipermutive. We prove that a Latin square generated by a CA is symmetric if and only if the local rule is $A_0(x + y) + B_0$, where x and y are the leftmost and rightmost variables, respectively, and $A_0 \neq 0$. Furthermore, we show that Latin squares generated by linear bipermutive CAs over the finite field $\mathbb{F}_q$ belong to the same isotopy class. Specifically, Latin squares generated by local rule functions of the form $f(x_1, x_2, \ldots, x_d) = A_0 + x_1 + A_2x_2 + \cdots + A_{d-1}x_{d-1} + x_d$, where $A_0, \ldots, A_{d-1} \in \mathbb{F}_q$ and d is the diameter, belong to the same isotopy class. Additionally, we prove that Latin squares in this isotopy class have the maximum number of intercalates when the CA is over $\mathbb{F}_2$ for any diameter d.

Computing the Character Table of a 2-local Maximal Subgroup ofthe Monster

Anthony Pisani
Monash University

The maximal subgroups of the Monster, the largest of the 26 sporadic simple groups, have recently been the subject of renewed interest. In 2023, Dietrich, Lee, and Popiel used Seysen's revolutionary `mmgroup` package for fast computations in the Monster to address three open cases for maximal subgroups, completing their classification. Burness and Hulpke computed several character tables later that year, while Popiel and Pisani calculated four outstanding sets of class fusions in 2024. In this talk, we discuss the use of `mmgroup` and Dietrich and Hulpke's hybrid group representation to compute the last remaining character table and class fusions of a maximal subgroup of the Monster.

A generating set for the commutator subgroup of a twisted right-angled Artin group

Vandit Trivedi
The Australian National University

A twisted right-angled Artin group (TRAAG) is defined by a mixed graph, which is a graph containing directed and possibly some undirected edges. The presence of directed edges distinguishes TRAAGs from right-angled Artin groups (RAAGs) by introducing a twisted relation between pairs of generators. We discuss the conditions on a mixed graph under which the commutator subgroup of a TRAAG is finitely generated and those under which the commutator subgroup has finite index in the TRAAG. We obtain a generating set for the commutator subgroup of a TRAAG and conjecture that it is minimal.

Construct Elusive Groups from Non-split Extensions

Binzhou Xia
The University of Melbourne

A finite transitive permutation group is *elusive* if it contains no derangements of prime order. These groups are closely related to a longstanding open problem in algebraic graph theory known as the Polycirculant Conjecture, which asserts that no elusive group is 2-closed. Existing constructions of elusive groups mostly arise from split extensions. In this talk, I'll give constructions of elusive groups via non-split extensions. This approach offers a key advantage: it allows the construction of elusive groups even when the acting group is not elusive. In particular, I'll construct elusive groups of new degrees, namely $p^{3k-4}(p+1)/2$ for each Mersenne prime $p \geq 7$ and integer $k \geq 2$, along with the first odd degree 225 and the first twice odd degree 450.

Perfect r-codes in circulant graphs

Yusuf Hafidh
The University of Melbourne

A perfect r-code in a graph G is an independent set C such that every vertex in $V(G) \setminus C$ is at distance at most r from a unique vertex in C. I will first discuss a "general" form for perfect 1-codes in circulant graphs (Cayley graphs over a cyclic group). Almost all known perfect 1-codes in circulant graphs are special cases of this general form. This form also characterizes several new families of circulant graphs that admit a perfect 1-code and classifies all such codes. I will also discuss perfect r-codes in circulant graphs with small degrees d, specifically for the parameters $(r, d) = (2, 3)$, $(2, 4)$, and $(2^l, 3)$.

Characterising groups by their element orders

Melissa Lee
Monash University

In recent years, there has been an explosion in the number of graphs defined from properties of finite groups. In this talk, we discuss one such genre of such graphs – those constructed from the orders of certain elements in the group. I will reveal surprisingly strong conditions on when a group can be characterised by such a small amount of information, along with some big problems that remain open in the area. Some of the work I will discuss is joint work with Tomasz Popiel, and with Kamilla Rekvényi.

Counting points on character varieties

Bailey Whitbread
The University of Sydney

This talk is about a pair of closely-related varieties called the multiplicative and additive character varieties. We study these varieties by counting their points over finite fields and prove the number of points is polynomial. Using GAP, we quickly and automatically compute these polynomials. We discuss striking features of these polynomials such as their non-negative, palindromic and unimodal coefficients.

Tuesday 18 of November

Session A - Room 3.08

Searching for 3-dimensional algebras

Adam Thomas

University of Warwick

Let $\mathfrak{g}$ be the Lie algebra of a simple algebraic group over an algebraically closed field of characteristic p. When $p = 0$ the celebrated Jacobson-Morozov Theorem promises that every non-zero nilpotent element of $\mathfrak{g}$ is contained in a simple 3-dimensional subalgebra of $\mathfrak{g}$ (an $\mathfrak{sl}_2$). This has been extended to odd primes but what about $p = 2$? There is still a unique 3-dimensional simple Lie algebra, known colloquially as fake $\mathfrak{sl}_2$, but there are other very sensible candidates like $\mathfrak{sl}_2$ and $\mathfrak{pgl}_2$. In this talk we will discuss recent joint work with David Stewart determining which nilpotent elements of $\mathfrak{g}$ live in subalgebras isomorphic to one of these three Lie algebras. There will be an abundance of concrete examples, calculations with small matrices and even some combinatorics.

Elementary abelian subgroups

Heiko Dietrich

Monash University

Knowing the elementary abelian p-subgroups of a finite group is a first step towards the classification of its p-radical and its maximal p-local subgroups. We describe some recent work that seeks to derive such a classification for the simple groups of exceptional type by first looking at their algebraic counterparts. This is joint work with Jianbei An and Alastair Litterick.

The AZ Diagram and Friends

Lavender Marshall

Monash University

Certain bordered Heegaard diagrams admit invariants which can be computed in terms of arc diagrams. I present a combinatorial proof of this together with some examples and open problems.

Discontinuous subgroups of $\mathsf{Aut}(\mathbb{S}^2)$ come in real-algebraic families with stable combinatorics

Alex Elzenaar
Monash University

Riemann surfaces are central objects in many fields. The Klein-Poincaré-Koebe uniformisation theorem (conj. late 1880s, proved 1907) shows that every Riemann surface is the set of orbits of a discontinuous group acting conformally on the disc. A generalisation by Bers-Maskit (early 1970s) shows that you can simultaneously put countably many of these discs-covering-surfaces onto the 2-sphere, and glue the corresponding groups together by HNN-extensions and amalgamated products, so that the new group acts conformally everywhere on the 2-sphere and acts discontinuously on the union of the embedded discs to produce all the Riemann surfaces with a single quotient operation. Marden observed (early/mid-1970s) that this should be the fundamental construction of the theory of infinite volume hyperbolic 3-manifolds. Since hyperbolic 3-manifolds are stable under deformation, the groups are stable under small algebraic deformations in $\mathsf{Aut}(\mathbb{S}^2) = \mathsf{PSL}(2, \mathbb{C})$. The region of stability is highly fractal. We exhaust it with explicit real semi-algebraic sets and in the process deduce a lot of information about the coarse geometry of the groups in the character variety of $\pi_1(M)$ for M a "sufficiently nice" (i.e. any M you naïvely draw on a piece of paper) hyperbolic 3-manifold. Focus in this talk will be placed on the (very applied) algebraic geometry that arises.

Computational Verification of the Sawin–Wood Conjectures for Distributions of Class Groups of C_7 Galois Number Fields Containing Root of Unity.

Mike Zhang
The University of Sydney

The class group is a central concept in number theory. As early as 1801, Gauss proposed three conjectures concerning class groups, and the third one remains an open question today. This reflects the deep complexity of their structure, which has led mathematicians to study the distribution of class groups instead. In this talk, I will explain the 1984 Cohen–Lenstra heuristics and the 2024 conjectures proposed by Sawin–Wood, along with Magma-based computational verifications supporting these distributional predictions.

A construction of magmas and related representation

Michal Botur
Palacký University in Olomouc

Given a magma $\mathbf{G} = (G, \cdot)$ and a biunary algebra $\mathbf{I} = (I, \lambda, \rho)$ (an algebra with two unary operations), we define a new magma $\mathbf{G}^{\mathbf{I}} = (G^I, \cdot)$ as follows: $(x \cdot y)(i) = x(\lambda i) \cdot y(\rho i)$ for any $x, y \in G^I$ and any $i \in I$. The main goal of the paper is to characterise the variety generated by all the magmas $\mathbf{G}^{\mathbf{I}}$ obtained in this way if $\mathbf{G}$ and $\mathbf{I}$ are taken from arbitrary (but fixed) varieties of magmas and biunary algebras, respectively.

Enriched Abstract Clones

John Power
Macquarie University

An algebraic signature together with equations is not an invariant of its algebras. For instance, there is more than one presentation of the notion of group. But each signature with equations generates an abstract clone, which is an invariant. A substantial part of universal algebra has been developed in terms of abstract, concrete and relational clones.

The definition of abstract clone is almost equivalent to the category theoretic definition of Lawvere theory, and clone theory, i.e., the body of mathematics surrounding clones, has been tweaked by Mike Behrisch in a way that is precisely consistent with Lawvere theories.

The definition of Lawvere theory generalises to that of enriched Lawvere theory, supporting, for instance, a notion of algebraic structure on posets that includes examples such as posets with finite meets.

Here, we show that a definition of abstract clone corresponding exactly to a Lawvere theory can be generalised to correspond to the definition of enriched Lawvere theory, thus opening the possibility of an extension of clone theory to an enriched setting. The key conceptual requirement is to replace the idea of a finite product of sets by a finite power of a set, with projections no longer explicit.

Quaternionic reflection groups: systems of imprimitivity

Don Taylor
The University of Sydney

In 1980 Arjeh Cohen published the classification of the finite quaternionic reflection groups. A notable example is the three-dimensional quaternionic representation of the double cover of the Hall–Janko group.

In recent years the connection with the generalised McKay correspondence and symplectic resolutions of quotient singularities has sparked renewed interest in these groups.

In this talk I will describe the structure of the rank 2 quaternionic reflection groups and their systems of imprimitivity. The results repair the classification, filling gaps and exhibiting duplications.

Counting points on algebraic curves and group-theoretic applications

Stephen Glasby
University of Western Australia

Graham Higman classified four families of finite 2-groups that have numerous applications in Group Theory. He named the groups after Michio Suzuki because of a connection with Suzuki's family of exceptional Lie-type simple groups. Computing the automorphism groups of Suzuki 2-groups raises challenging problems related to counting points on certain algebraic curves. I will give some context and applications of group-related polynomial equations $p(x, y) = 0$ for (x, y) in a given finite field. This is joint with Peter Müller from the University of Würzburg.

Tuesday 18 of November

Session B - Room 3.09

Colored blob algebras

Mengfan Lyu
Western Sydney University

In this talk, I will introduce the colored blob algebra $CB_{r,n}$, a multi-parameter generalization of the classical Temperley-Lieb algebra that incorporates blobs of $r-1$ distinct colors. I will begin by defining its diagrammatic basis and the relations that govern it.

The main focus will be on establishing the algebraic foundations of $CB_{r,n}$. In particular, I will show that it arises as a quotient of the cyclotomic Hecke algebra $H_{r,1,n}$. This result extends the well-known relationship between the original blob algebra and the Hecke algebra of type B, situating the colored blob algebra naturally within the broader family of cyclotomic Hecke algebra quotients.

I will also demonstrate that $CB_{r,n}$ is a cellular algebra. This structural property ensures a tractable representation theory and facilitates the explicit construction of its simple modules. The framework developed here opens up new directions for research in diagram algebras, representation theory, and related areas.

Runner removal and Stingray pattern.

Tao Qin
The University of Sydney

The runner removal theorems describe the independence of decomposition numbers on the quantum characteristic e; we give a new proof using alcoves. Along with the proof, we find an interesting Stingray pattern of the dominant weight lattice of sl_3.

On representation zeta functions of Sp_4 over compact discrete valuation rings

Seethalakshmi Kayanattath
Australian National University

Let G be a group, and for a natural number n, let $r(n, G)$ denote the number of n-dimensional complex irreducible representations of G up to isomorphism. If G is topological or algebraic, we assume further that the representations are continuous or algebraic, respectively. We can gather insight into the structural properties of the group by analysing the asymptotics of the sequence $r(n, G)$ by encoding it in a generating function called the representation zeta function.

In this talk, we will focus on the representation zeta function of the rank-2 symplectic group over a compact discrete valuation ring with finite residue field of odd characteristic. Using Clifford theory, the irreducible representations of the finite quotients of this group can be organised in terms of the adjoint orbits of the symplectic group acting on the underlying abelian group of its Lie algebra over the finite residue field. We begin by classifying these adjoint orbits into regular, decomposable, 2-primary and nilpotent types. We study the part of the representation zeta function encoding the representations corresponding to each type of orbit separately. The difficulty in the computation varies significantly with the type of the orbit.

Semisimplifying categorical Heisenberg actions, with applications to modular representation theory of symmetric groups

Finn Klein
The University of Sydney

The Duflo-Serganova functor is an indispensable tool in the representation theory of Lie superalgebras. In joint work with Hone-Pauwels-Sherman-Yacobi-Zhang, we introduce OTI functors, which mirror this construction in the setting of categorical Heisenberg actions. Applied to classical categorifications of Fock space, OTI functors yield a global construction of periodic equivalences which have been studied by Henke-Koenig and Harman, providing a conceptual source for these phenomena. In the talk we will motivate and introduce OTI functors and demonstrate applications to the representation theory of symmetric groups.

Every Generalized Monoid Ring is Definable by a D-structure

Barry Gardner

A *generalized monoid ring* for a ring R with identity and a monoid G is a ring S with identity such that R is a subring with the same identity and S has a multiplicative submonoid $G' \cong G$ which is a basis for S as a free left unital R-module. Examples include monoid rings, skew polynomial rings (a.k.a. Ore extensions) and skew group rings. One way of constructing such a ring from R and G is by means of a *D-structure*, a collection of self-maps $\sigma_{x,y}$ of R labelled by the elements of $G \times G$ satisfying a long list of mostly natural-looking conditions, in effect defining a new multiplication on the ordinary monoid ring given by

$$ax \cdot by = a \sum_z \sigma_{x,z}(b) zy.$$

It turns out that every generalized monoid ring can be defined in this way.

Finite rings and sum-product phenomenon

Alexander Fish

The University of Sydney

Let R be a finite ring and a $E \subset R$ a subset of density $\alpha > 0$. Is it true that the product-set $(E - E) \cdot (E - E)$ necessarily contains a subring of R whose index depends only on α? We will present a short proof showing that this is indeed the case when $R = \mathbb{Z}/N\mathbb{Z}$.

Convolution rings

Stefan Veldsman

Nelson Mandela University & La Trobe University

Convolution types and the corresponding induced convolution rings were introduced earlier for two reasons. Firstly, they provide a unified and convenient framework to talk about several apparent diverse ring constructions. Secondly, the properties of the parameters of a convolution type may explain why different constructions over the same underlying ring may have different properties. Many examples will be given. These will include the arithmetic convolution rings covering, amongst others, the Cauchy product and the Dirichlet product.

Lambda lengths in link complements

Joshua Howie
Monash University

Lambda lengths measure the complex lengths of certain geodesic segments in hyperbolic link complements. Associated to each hyperbolic 3-manifold is a number field called the trace field. Mathews showed that there is an $SL_2(\mathbb{C})$-equivariant correspondence between nonzero spinors and decorated horospheres. Using this correspondence, and studying the action of link groups on sets of decorated horospheres, we show that the set of lambda lengths in several arithmetic link complements is the ring of integers in their trace field, up to multiplication by a unit. We also consider some non-arithmetic cases. Joint work with Nicola Harif, Connie On Yu Hui, Dionne Ibarra, Daniel Mathews, and Lecheng Su.
